English

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ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2025-2026)

Test Pattern

NEET (UG)

MINOR

07-09-2025

PRE-MEDICAL : LEADER & ACHIEVER COURSE (SRG)

Test Booklet Code

E1

This Booklet contains **28** pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
9. No candidate, without special permission of the Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
13. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
14. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

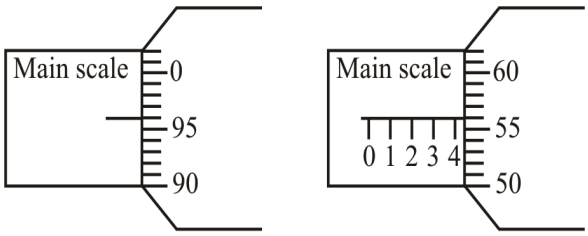
Candidate's Signature : _____ **Invigilator's Signature :** _____

Your Target is to secure Good Rank in Pre-Medical 2026

SUBJECT : PHYSICS

Topic : BASIC MATHS,UNIT & DIMENSION,ERROR MEASUREMENT, KINEMATICS (UP TO 1D), ELECTROSTATICS COMPLETE.

1. Two full turns of Screw gauge cover a distance of 1 mm on main scale. The total number of circular scale division are 100. The measured value of thickness of sheet will be :-



Screw Gauge

(1) 4.555 mm

(2) 4.295 mm

(3) 2.300 mm

(4) 2.245 mm

2. Match the dimensional formula. (Here y represents distance, t represents time and [A] represents dimension of A.

	Column-I		Column-II
(A)	$\left[\frac{dy}{dt}\right]$	(P)	$M^0L^0T^{-2}$
(B)	$[\alpha]$ in equation, $y = A\cos(\alpha^2t^2)$	(Q)	$M^1L^1T^1$
(C)	$\left[\log\left(e^{-\alpha^2t^2}\right)\right]$	(R)	$M^0L^0T^0$
(D)	$[\alpha]$ in equation, $y_2 = y_1(1 + \alpha(t_2 - t_1))$	(S)	$M^0L^1T^{-1}$
		(T)	$M^0L^0T^{-1}$

(1) A-S, B-T, C-R, D-P

(2) A-S, B-T, C-R, D-T

(3) A-S, B-T, C-P, D-Q

(4) A-S, B-P, C-Q, D-T

3. Consider the following statements:

(a) The plot of systematic error versus number of observations in any experiment is a straight line parallel to number of observations axis.

(b) Systematic errors are either positive or negative.

(c) Random errors occur irregularly and hence are random w.r.t sign and size.

Choose the correct statements.

(1) (a) and (b) only (2) (b) only

(3) (b) and (c) only (4) (a), (b) and (c)

4. According to Vanderwall's equation pressure(P) volume(V) and Temperature(T) are related as $\left(P + \frac{a}{V^2}\right)(V - b) = RT$ (for 1 mole of gas). Then dimensions of $\frac{ab}{V^2}$ is equivalent to :-

(1) Force (2) Power

(3) Energy (4) Pressure

5. Two resistance are measured in Ohm and is given as :-

$R_1 = 3\Omega \pm 1\%$ & $R_2 = 6\Omega \pm 2\%$

When they are connected in parallel, the percentage error in equivalent resistance is :-

(1) 3% (2) 4.5%

(3) 0.67% (4) 1.33%

6. The period of oscillation of simple pendulum is $T = 2\pi\sqrt{L/g}$. Measured value of L is 20.0 cm known to 1mm accuracy and time for 100 oscillations of the pendulum is found to be 90 s using a wrist watch of 1 s resolution. What is the accuracy in the determination of g to nearest integer?

(1) 1% (2) 3% (3) 6% (4) 9%

7. A physical quantities given by the relation $P^3 = \frac{Q^{1/3}}{5RS^2}$, if the percentage error are in P, Q, & S is 2%, 3% & 1%, than find out the % error in R ?

(1) 12% (2) 13% (3) 9% (4) 8.6%

8. If $P = \int_{x_1}^{x_2} \frac{dx}{Q - x^2}$, where x is position, P and Q are unknown quantities, then dimensions of PQ are :-

(1) [L] (2) $[L^2]$

(3) $[L^3]$ (4) $[L^{-1}]$

9. A ball is thrown upwards from a planet such that distance travelled in the 4th and 5th second are equal. If the maximum height attained by the ball is 32m, choose the correct statement(s).

(A) Acceleration due to gravity of the planet is 2 m/s^2.

(B) Acceleration due to gravity of the planet is 4 m/s^2.

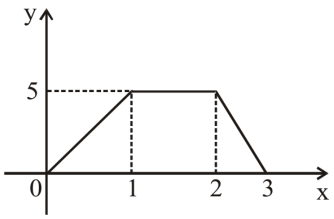
(C) Distance travelled by the ball in 4th second is 2m.

(D) Time taken for the ball to strike ground again is 10 s.

(1) A, C (2) B, C

(3) B, C, D (4) None of these

10. If



than match the columns :-

	Column (I)		Column (II)
(A)	$\int_0^3 y\,dx$	(P)	0
(B)	$\left(\frac{dy}{dx}\right)_{x=0.5}$	(Q)	10
(C)	$\left(\frac{d^2y}{dx^2}\right)_{x=2.5}$	(R)	5

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – Q, B – R, C – Q

(4) A – P, B – Q, C – R

11. Which of the following are correct :-

(a) Average speed of a particle in a given time interval is never less than the magnitude of average velocity.

(b) It is possible to have a situation in which $\left|\frac{d\vec{v}}{dt}\right| \neq 0$ but $\frac{d}{dt}|\vec{v}| = 0$

(c) The average velocity of a particle is zero in a time interval. It is possible than the instantaneous velocity is never zero in the interval

(d) The average velocity of a particle moving on a straight line is zero in a time interval. It is possible that the velocity is never zero in the interval. (Infinite accelerations are not allowed)

(1) a,b,c (2) a,b,d (3) b,c (4) a only

12. Velocity (in m/s) of a particle moving in a straight line is given by $v = (t^2 - 2t + 1)$:-

	List-I		List-II
(P)	Velocity (in m/s) of particle at t = 3 sec is	(1)	1
(Q)	Acceleration (in m/s^2) of particle at t = 2 sec is	(2)	2
(R)	Time when particle is at rest is (in second)	(3)	3
(S)	Magnitude of average acceleration of particle in first one second is (in m/s^2)	(4)	4

Codes:-

	P	Q	R	S
(1)	3	2	4	1
(2)	4	3	2	1
(3)	4	2	1	1
(4)	1	4	3	2

13.

COLUMN – I		COLUMN – II	
A	Area under velocity time graph	P	Change in velocity
B	Area under acceleration time graph	Q	Displacement
C	Slope of velocity time graph	R	Acceleration.
D	Slope of displacement time graph	S	Velocity

(1) A- R; B – P; C – Q; D – S

(2) A- Q; B – P; C – R; D – S

(3) A- Q; B – P; C – S; D – R

(4) A- S; B – R; C – P; D – Q

14. Find the appropriate option from column-II for column-I, if position of a particle is described by $x = 3 + 8t - 4t^2$.

	Column-I		Column-II
(a)	Initial acceleration	i	−16 unit
(b)	Velocity at end of 3s	ii	8 unit
(c)	distance travelled in 2s	iii	4 unit
(d)	displacement at 1s	iv	−8 unit

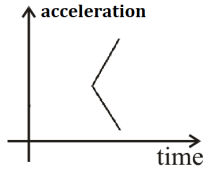
(1) (a)-iv ; (b)- (i) ; (c)- (iii) ; (d)- ii

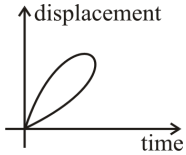
(2) (a)-iv ; (b)- (i) ; (c)- (ii) ; (d)- iii

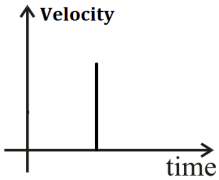
(3) (a)-ii ; (b)- (iii) ; (c)- (iv) ; (d)- i

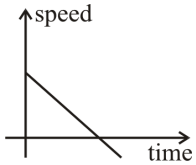
(4) (a)-ii ; (b)- (iii) ; (c)- (i) ; (d)- iv

15. Which of the graphs is realistically not possible :-

(P) 

(Q) 

(R) 

(S) 

(1) P, Q, R, S (2) Q, R, S

(3) P, Q, R (4) Q, S

16. The displacement x of a particle is $x = \frac{a}{b}\left[1 - e^{-bt}\right]$ where a & b are positive constants, then velocity and acceleration at t = 0 is –

(1) a & – ab

(2) – a & ab

(3) – a & – ab

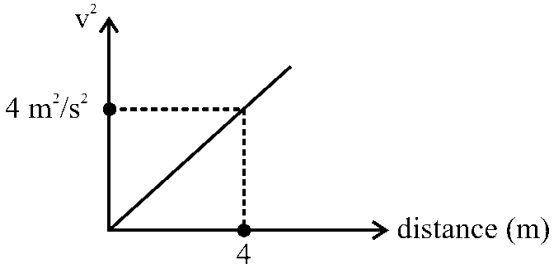
(4) a & ab

17. If a ball is thrown vertically upwards with speed u, the distance covered during the last t seconds of its ascent is :-

(1) $\frac{1}{2}gt^2$ (2) $ut - \frac{1}{2}gt^2$

(3) (u – gt)t (4) ut g

18. A particle is moving on a straight line such that square of its velocity is directly proportional to distance as shown in graph. What is the acceleration of the particle ?



(1) $\frac{1}{4}m/s^2$ (2) 1 m/s^2

(3) 2 m/s^2 (4) $\frac{1}{2}m/s^2$

19. A car is moving with speed 15 m/s. Suddenly the driver sees an obstruction on the road and takes 0.4 sec to apply the brake, which cause a deacceleration of 5 m/s^2. Then distance travelled by car before stopping-

(1) 22.5 m (2) 24.5 m

(3) 26.5 m (4) 28.5 m

20. A particles moving along a straight line has a velocity v m s^{-1}, when it cleared a distance of x m. These two are connected by the relation $v = \sqrt{49 + x}$. When its velocity is 1 m s^{-1}, its acceleration is :-

(1) 2 m s^{-2}

(2) 7 m s^{-2}

(3) 1 m s^{-2}

(4) 0.5 m s^{-2}

21. A particle moving with uniform acceleration has velocity 6 m/s at a distance of 5 m from the initial position. After moving another 7 m, the velocity becomes 8 m/sec. The initial velocity and acceleration of the particle are :-

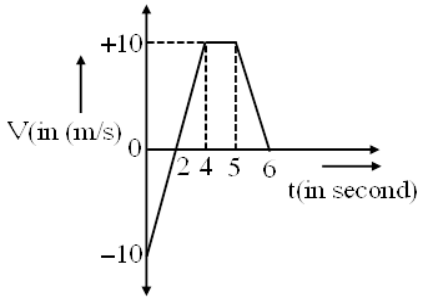
(1) 2 m/s, 4 m/s^2 (2) 4 m/s, 2 m/s^2

(3) 4 m/s, 4 m/s^2 (4) 6 m/s, 1 m/s^2

22. The engine of a motorcycle can produce a maximum acceleration 5 m/s^2. Its brakes can produce a maximum retardation 10 m/s^2. If motorcyclist start from point A and reach at point B. What is the minimum time in which it can cover if distance between A and B is 1.5 km. (Given : that motorcycle comes to rest at B)

(1) 30 sec (2) 15 sec (3) 10 sec (4) 5 sec

23. For the velocity time (v-t) graph shown in figure, the incorrect statement is :-



(1) The average velocity for the entire journey is 2.5 m/s

(2) The average acceleration from 1s to 4s is 5 m/s^2

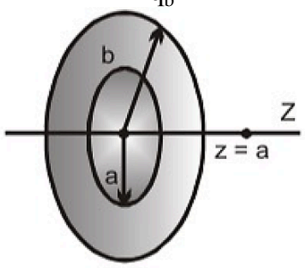
(3) The average speed for the first 4s is zero

(4) The acceleration at t = 3s is 5 m/s^2

24. A solid sphere of radius R_1 and volume charge density $\rho = \frac{\rho_0}{r}$ is enclosed by a hollow sphere of radius R_2 with negative surface charge density σ, such that the total charge in system is zero, Here ρ_0 is +ve constant and r is distance from centre of sphere. The ratio $\frac{R_2}{R_1}$ is :

(1) $\frac{\sigma}{\rho_0}$ (2) $\sqrt{\frac{2\sigma}{\rho_0}}$ (3) $\sqrt{\frac{\rho_0}{2\sigma}}$ (4) $\frac{\rho_0}{\sigma}$

25. Two concentric rings one of radius a and the other of radius b = 2a have charges q_a and a_b respectively as shown in figure. A charged particle placed on the axis at z = a is in equilibrium, if the ratio $\frac{q_a}{q_b}$ is equal to _____.



(1) $-\left(\frac{2}{5}\right)^{\frac{3}{2}}$ (2) $\left(\frac{2}{5}\right)^{\frac{3}{2}}$

(3) $\left(\frac{2}{5}\right)^{\frac{2}{3}}$ (4) $-\left(\frac{2}{5}\right)^{-\frac{3}{2}}$

26. A particle of charge $-q$ and mass m moves in a circular orbit of radius r about a fixed charge +Q. The relation between the radius of the orbit r and the time period T is :

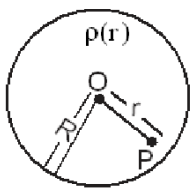
(1) $r = \frac{Qq}{16\pi^2 \in_0 m} T^2$ (2) $r^3 = \frac{Qq}{16\pi^3 \in_0 m} T^2$

(3) $r^2 = \frac{Qq}{16\pi^3 \in_0 m} T^3$ (4) $r^2 = \frac{Qq}{4\pi^3 \in_0 m} T^3$

27. A spherically symmetric charge distribution is considered with charge density varying as

$$\rho(r) = \begin{cases} \rho_0\left(\frac{3}{4} - \frac{r}{R}\right) & \text{for } r \leqslant R \\ \text{Zero} & \text{for } r > R \end{cases}$$

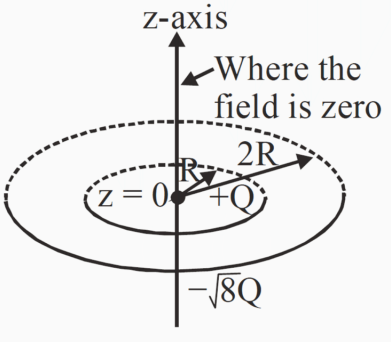
Where, r(r < R) is the distance from the centre O (as shown in figure). The electric field at point P will be :



(1) $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{3}{4} - \frac{r}{R}\right)$ (2) $\frac{\rho_0 r}{3\,\varepsilon_0}\left(\frac{3}{4} - \frac{r}{R}\right)$

(3) $\frac{\rho_0 r}{4\varepsilon_0}\left(1 - \frac{r}{R}\right)$ (4) $\frac{\rho_0 r}{5\,\varepsilon_0}\left(1 - \frac{r}{R}\right)$

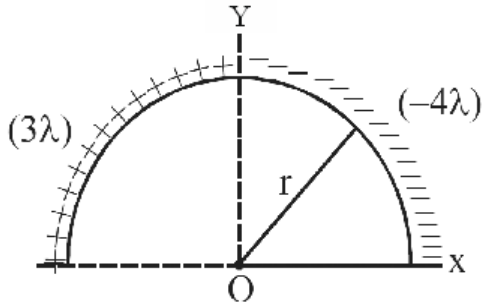
28. Two concentric rings, one of radius R and total charge +Q and the second of radius 2R and total charge $-\sqrt{8}Q$, lie in x-y plane (i.e., z = 0 plane). The common centre of rings lies at origin and the common axis coincides with z-axis. The charge is uniformly distributed on both rings. At what distance from origin is the net electric field on z-axis zero?



(1) $\frac{R}{2}$ (2) $\frac{R}{\sqrt{2}}$

(3) $\frac{R}{2\sqrt{2}}$ (4) $\sqrt{2}R$

29. A semicircular arc placed in x-y plane is charged as shown in figure. The electric field at centre 'O' is.



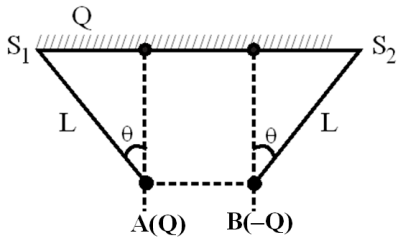
(1) $\frac{5\sqrt{2}\lambda}{4\pi\varepsilon_0 r}$ above x direction

(2) $\frac{5\sqrt{2}\lambda}{4\pi\varepsilon_0 r}$ below x direction

(3) $\frac{5\sqrt{2}\lambda}{4\pi\varepsilon_0 r}$ at 45° with x-axis

(4) $\frac{5\sqrt{2}\lambda}{4\pi\varepsilon_0 r}$ along x direction

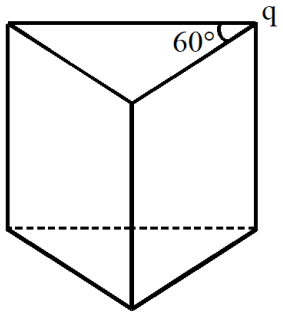
30. Two balls A and B of same mass (M) and charges Q, −Q are suspended by two strings of same length from two different suspension points S_1 and S_2 if $S_1S_2 = 3x$ and AB = x then tension in the string will be.



(1) $\frac{Q^2L}{4\pi\varepsilon_0 x^3}$ (2) $\frac{QL}{4\pi\varepsilon_0 x^3}$

(3) $\frac{KQL}{x^2}$ (4) None of these

31. A charge q is placed at a vertex of the given geometry of angle 60°. Then, the flux coming out from this geometry will be :



(1) $\frac{q}{6\epsilon_0}$ (2) $\frac{q}{16\epsilon_0}$

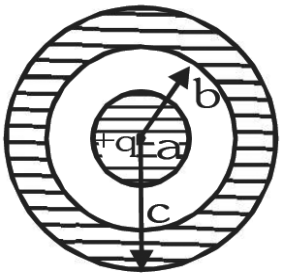
(3) $\frac{q}{8\epsilon_0}$ (4) $\frac{q}{12\epsilon_0}$

32. Two isolated metallic solid spheres of radii R and 2R are charged such that both have same charge density σ. The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ'. The ratio $\frac{\sigma'}{\sigma}$ is :

(1) $\frac{9}{4}$ (2) $\frac{4}{3}$

(3) $\frac{5}{3}$ (4) $\frac{5}{6}$

33. A non conducting sphere of radius 'a' has a net charge '+q' uniformly distributed throughout its volume. A hollow spherical conductor having inner and outer, radii 'b' and 'c' and net charge '−q' is concentric with the sphere (see the figure). Read the following statements



(i) The electric field at a distance r from the centre of the sphere $= \frac{1}{4\pi\varepsilon_0}\frac{qr}{a^3}$ for r < a

(ii) The electric field at distance r from the centre of the sphere for a < r < b = 0

(iii) The electric field at distance r from the centre of the sphere for b < r < c = 0

(iv) The charge on the inner surface of the hollow spherie = −q

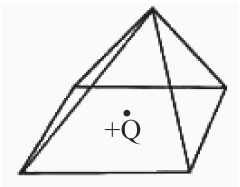
(v) The charge on the outer surface of the hollow sphere = +q

Which of the above statements are true ?

(1) (i), (ii) and (v) (2) (i), (iii) and (iv)

(3) (ii), (iii) and (iv) (4) (ii), (iii) and (v)

34. A point charge +Q is positioned at the centre of the base of square pyramid as shown. The flux through one of the four identical upper faces of the pyramid is :



(1) $\frac{Q}{16\in_0}$ (2) $\frac{Q}{4\in_0}$

(3) $\frac{Q}{8\in_0}$ (4) None of these

35. Three point charges q_1, q_2 and q_3 are taken such that when q_1 and q_2 are placed close together to form a single point charge, the force on q_3 at distance L from this combination is a repulsion of 2 units in magnitude. When q_2 and q_3 are so combined the force on q_1 at distance L is an attractive force of magnitude 4 units. Also q_3 and q_1 when combined exert an attractive force on q_2 of magnitude 18 unit at same distance L. The algebraic ratio of charges q_1, q_2 and q_3 is

(1) 1: 2: 3

(2) 2: −3 : 4

(3) 4 : −3 : 1

(4) 4: −3 : 2

36. The electric field in a certain region is acting radially outward and is given by E = Ar. The charge contained in a sphere of radius 'a' centred at the origin of the field, will be given by :

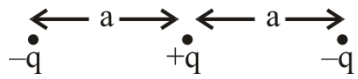
(1) $A\,\varepsilon_0\,a^2$

(2) $4\,\pi\varepsilon_0\,Aa^3$

(3) $\varepsilon_0\,Aa^3$

(4) $4\,\pi\varepsilon_0\,Aa^2$

37. Three points charges shown in the figure lie along a straight line. The energy needed to exchange the position of the central +ve charge with one of the negative charges is :-



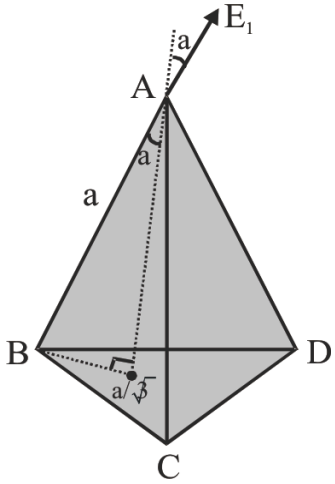
(1) $\frac{q^2}{8\pi\in_0 a}$

(2) $\frac{3q^2}{8\pi\in_0 a}$

(3) $\frac{q^2}{4\pi\in_0 a}$

(4) $\frac{-q^2}{8\pi\in_0 a}$

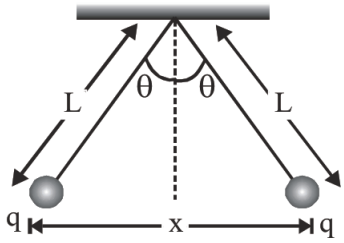
38. A,B,C,D are the corners of a regular tetrahedron of side a. Three identical charges each Q are fixed at three corners B, C and D. The magnitude of electric field intensity at the corner A is :-



(1) $\frac{\sqrt{6}Q}{4\pi\in_0 a^2}$ (2) $\frac{2\sqrt{3}Q}{2\pi\in_0 a^2}$

(3) $\frac{\sqrt{6}Q}{2\pi\in_0 a^2}$ (4) $\frac{2\sqrt{3}Q}{4\pi\in_0 a^2}$

39. In the given figure two tiny conducting balls of identical mass m and identical charge q hang from non-conducting threads of equal length L. Assume that θ is so small that tan θ ≈ sinθ, then for equilibrium x is equal to :-



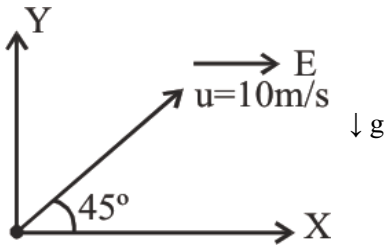
(1) $\left(\frac{q^2L}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(2) $\left(\frac{qL^2}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(3) $\left(\frac{q^2L^2}{4\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(4) $\left(\frac{q^2L}{4\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

40. A charged particle of mass m = 2 kg and charge 1 mC is thrown from a horizontal ground at an angle of 45º with horizontal with a speed of 10 m/s. In space a horizontal electric field E = 2×10^7 N/C also exists along with verticle down g = 10 m/s^2 gravitational field. Find the range of the projectile.

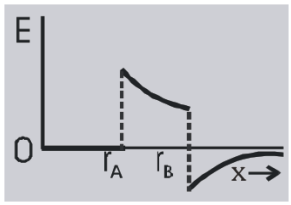


(1) 5 m (2) 30 m

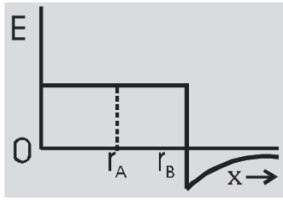
(3) 10 m (4) 20 m

41. Two concentric conducting thin spherical shells A, and B having radii r_A and r_B ($r_B > r_A$) are charged with charges Q_A and $-Q_B$ ($|Q_B| > |Q_A|$). The electric field along a line, (passing through the centre) varies with distance x from the centre as :-

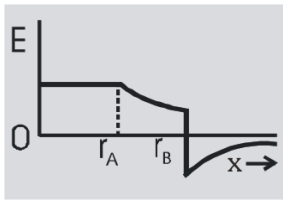
(1)



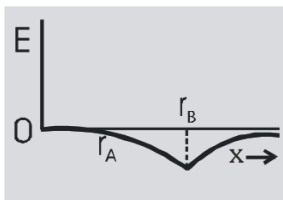
(2)



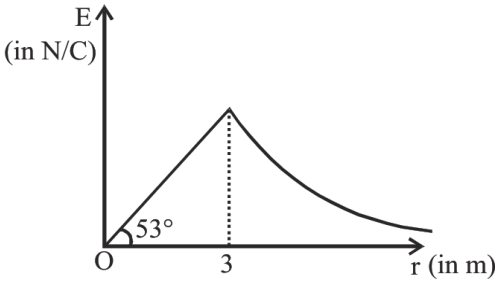
(3)



(4)



42. We have a solid non conducting uniformly charged sphere. Its electric field variation with distance from centre is shown in graph then the net charge inside the sphere (in S.I. units) is :-



(1) $144\ \pi\in_0$ (2) $100\ \pi\in_0$

(3) $32\ \pi\in_0$ (4) $48\ \pi\in_0$

43. A rod of length L has a total charge Q distributed uniformly along its length. It is bent in the shape of a semicircle. Find the magnitude of the electric field at the centre of curvature of the semicircle.

(1) $Q/2\in_0 L^2$ (2) $Q/\in_0 L^2$

(3) $\frac{2Q}{\in_0 L^2}$ (4) Zero

44. Dielectric constant of pure water is 81. It's permittivity will be (SI unit)

(1) 7.17×10^{-10}

(2) 8.86×10^{-12}

(3) 1.02×10^{-13}

(4) None

45. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 30º with each other. When suspended in a liquid of density 0.8 g cm^{-3}, the angle remains the same. If density of the material of the sphere is 1.6 g/cm^3 , the dielectric constant of the liquid is:-

(1) 4

(2) 3

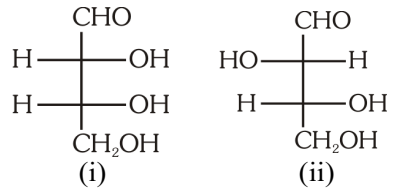
(3) 2

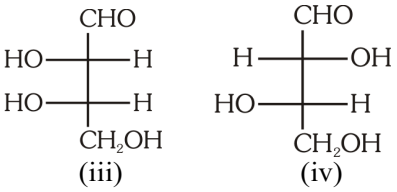
(4) 1

SUBJECT : CHEMISTRY

Topic : ATOMIC STRUCTURE, MOLE CONCEPT , REDOX REACTION, IUPAC , ISOMERISM COMPLETE, PERIODIC TABLE , CHEMICAL BONDING (UP TO BB-3).

46. The correct corresponding order names of four aldoses with configuration given below





respectively, is :-

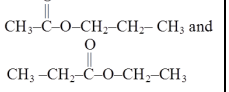
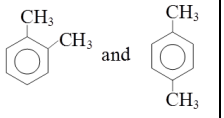
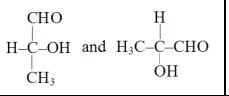
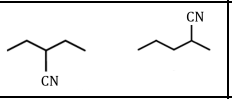
(1) L-erythrose, L-threose, D-erythrose, D-threose

(2) D-erythrose, D-threose, L-erythrose, L-threose

(3) L-erythrose, L-threose, L-erythrose, D-threose

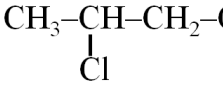
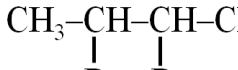
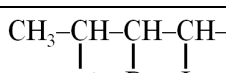
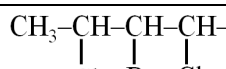
(4) D-threose, D-erythrose, L-threose, L-erythrose

47. Match list I with List II and select the correct answer from the codes given below the lists :

List-I		List-II	
(A)		(P)	Enantiomer
(B)		(Q)	Position isomers
(C)		(R)	Metamers
(D)		(S)	Chain Isomers

	(A)	(B)	(C)	(D)
(1)	R	Q	P	S
(2)	R	Q	S	P
(3)	P	Q	S	R
(4)	Q	R	P	S

48.

	Column-I (Structure)		Column-II (Number of meso & optical isomers)
A.		1.	M = 2, OI = 4
B.		2.	M = 0, OI = 2
C.		3.	M = 0, OI = 8
D.		4.	M = 1, OI = 3

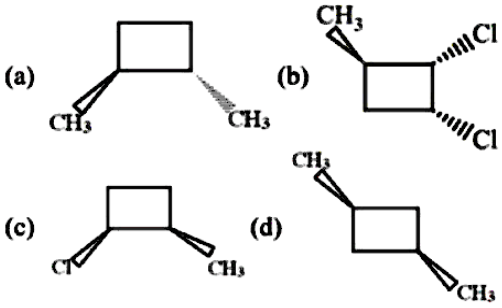
(1) (A−2), (B−4), (C−3), (D−1)

(2) (A−1), (B−2), (C−3), (D−4)

(3) (A−2), (B−3), (C−4), (D−1)

(4) (A−3), (B−4), (C−1), (D−2)

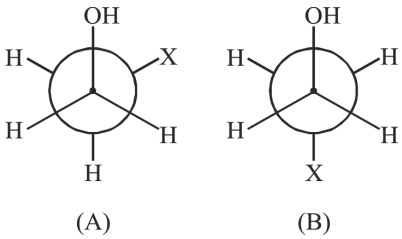
49. Which of the following are chiral ?



(1) a, b, c (2) a, b, d

(3) b, c, d (4) a, c, d

50.



(A) (B)

A will be more stable than B if X is :

(1) $-CH_3$ (2) $-C_2H_5$

(3) $-F$ (4) $-Br$

51. **Assertion :** Staggered conformation of ethane is most stable while eclipsed conformation is least stable.

Reason : Staggered form has the least torsional strain and the eclipsed form has the maximum torsional strain

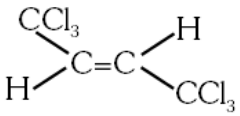
(1) Assertion is correct, reason is correct, reason is a correct explanation for assertion

(2) Assertion is correct, reason is correct, reasion is not a correct explanation for assertion

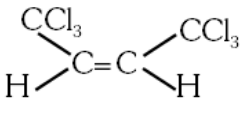
(3) Assertion is correct, reason is incorrect.

(4) Assertion is incorrect, reason is correct.

52. Which is true for the following isomeric forms I & II respectively :



(I)



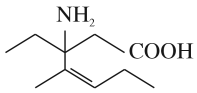
(II)

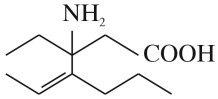
	Dipole moment	Boiling point	Melting point	Stability
(1)	I > II	I > II	II > I	I > II
(2)	II > I	II > I	II > I	II > I
(3)	I > II	I > II	I > II	I > II
(4)	II > I	II > I	I > II	I > II

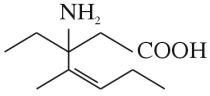
53. How many aliphatic carbonyl compound are possible by $C_5H_{10}O$:-

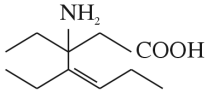
(1) 8 (2) 5 (3) 6 (4) 7

54. Structure of the compound whose IUPAC name is 3-Amino-3, 4-diethylhept-4-enoic acid

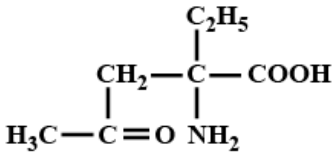
(1) 

(2) 

(3) 

(4) 

55. The IUPAC name of compound is :-



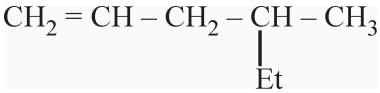
(1) 2-Oxo-4-amino-4-ethyl pentanoic acid

(2) 2-Amino-4-oxo-2-ethyl pentanoic acid

(3) 3-Amino-3-carboxy-5-hexanone

(4) 2-Amino-2-ethyl-4-oxo pentanoic acid

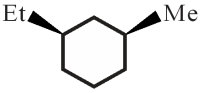
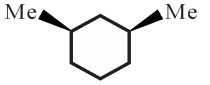
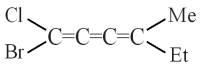
56. The correct IUPAC name of the compound



(1) 4-Ethylpent-1-ene (2) 2-Ethylpent-4-ene

(3) 4-Methylhex-1-ene (4) 3-Methylhex-1-ene

57.

	Column-I		Column-II
A.		p.	Compound is achiral
B.		q.	Compound is dissymetric
C.		r.	Compound has chiral carbons
D.		s.	Compound has diastereomer/s
		t.	Compound has plane of symmetry

(1) A−r,s; B−p,q,s,t; C−p,s,t; D−q

(2) A−p,r,s; B−p,r,t; C−p,q,t; D−r

(3) A−r,s; B−p,r,s,t; C−p,r,t; D−s

(4) A−q,r,s; B−p,r,s,t; C−p,s,t; D−q

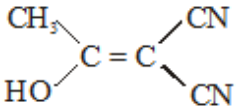
58. The correct statement for Isooctane is −

(1) It has seven carbons in its principal carbon chain.

(2) It has five secondary carbons.

(3) It has one quarternary carbon and one tertiary carbon.

(4) It has ten primary hydrogens.

59. In  compound no. of σ & no. of π bond is respectively

(1) 4, 3 (2) 5, 5

(3) 3, 4 (4) 11, 5

60. ***Assertion :-*** A compound whose molecule has D-configuration will always be dextrorotatory.

Reason :- Compound having D-configuration is always rotate the plane of plane polarised light in clockwise direction.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** and **Reason** both are false

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

61. Given, uncertainty in momentum of an electron 3 times of uncertainty in position. Find uncertainty in velocity of that electrons.

(1) $\frac{1}{4m}\sqrt{\frac{h}{3\pi}}$ (2) $\frac{1}{2m}\sqrt{\frac{3h}{\pi}}$

(3) $\frac{1}{3m}\sqrt{\frac{4\pi}{h}}$ (4) $\frac{1}{3m}\sqrt{\frac{h}{\pi}}$

62. For p orbital electron, orbital angular momentum is

(1) $\frac{\sqrt{6}h}{2\pi}$ (2) $\frac{\sqrt{2}h}{2\pi}$

(3) $\frac{h}{2\pi}$ (4) Zero

63. The wave number of electromagnetic radiation emitted during the transition of electron in between the two energy levels of Li^{+2} ion whose principal quantum number sum is 4 and difference is 2 is :-

(1) 3.5 R_H (2) 4 R_H

(3) 8 R_H (4) $\frac{8}{9}$ R_H

64. **Statement (1) :** d_{z^2} orbital has higher energy than that of d_{xy} or d_{yz} or d_{xz} of same shell in absence of electric and magnetic field.

Statement (2) : Total number of orbitals associated with the principal quantum number n = 3 is 9.

Statement (3) : The shapes of all five d-orbitals are similar to each other.

Choose the correct option where T is true and F is false -

(1) TTF (2) TFT (3) FTF (4) FFF

65. In the hydrogen spectrum which of the following has maximum wavelength.

(1) First line of lymen series

(2) First line of Balmer series

(3) Last line of Paschen series

(4) Last line of Brackett series

66. Find the weight of sample of $KClO_3$ to produce 96 g O_2 and percent yield reaction is 80% and purity of sample is 75%.

(1) 408.33 g (2) 490 g

(3) 1225 g (4) 544.44 g

67. Vapour density of mixture containing NO_2 and N_2O_4 is 27.6 then Calculate weight % of NO_2 in mixture

(1) 20 (2) 80 (3) 33.33 (4) 66.66

68. Match the column :-

	Column-I		Column-II
(A)	8.8 g of CO_2	(P)	36.132×10^{23} atoms of oxygen
(B)	5.6 L of CO_2 at STP	(Q)	12.044×10^{22} molecule
(C)	96 g of O_2	(R)	0.25 mol
(D)	49 g of H_2SO_4	(S)	0.5 mol

(1) A⟶Q, B⟶R, C⟶S, D⟶P

(2) A⟶Q, B⟶R, C⟶P, D⟶S

(3) A⟶P, B⟶R, C⟶S, D⟶Q

(4) A⟶P, B⟶R, C⟶Q, D⟶S

69. Chlorine is prepared in the laboratory by treating manganese dioxide (MnO_2) with aqueous hydrochloric acid according to the reaction
$4HCl_{(aq.)} + MnO_{2(s)} \longrightarrow 2H_2O_{(\ell)} + MnCl_{2(aq)} + Cl_{2(g)}$
How many grams of HCl react with 5.0 g of MnO_2 ? ($MnO_2 = 87$)

(1) 5g (2) 0.23g

(3) 3g (4) 8.4g

70. Sulphur forms two chlorides S_2Cl_2 and SCl_2. The equivalent mass of sulphur in SCl_2 is :-

(1) 8 (2) 16

(3) 64.8 (4) 32

71. The crystalline salt $Na_2SO_4.xH_2O$ on heating loses 50.35% of its mass and become anhydrous. The formula of crystalline salt is.

(1) $Na_2SO_4.10H_2O$ (2) $Na_2SO_4.8H_2O$

(3) $Na_2SO_4.5H_2O$ (4) $Na_2SO_4.H_2O$

72. 50g sample of $CaCO_3$ on heating gives 17.6g of CO_2. Calculate % purity of sample ?

(1) 80% (2) 20% (3) 30% (4) 50%

73. Oxidation number of carbon in C_3O_2 is

(1) $-4/3$ (2) $+4/3$

(3) $+2/3$ (4) $-2/3$

74. For the redox reaction,
$MnO_4^- + C_2O_4^{2-} + H^+ \longrightarrow Mn^{2+} + CO_2 + H_2O$
the correct coefficients of MnO_4^-, $C_2O_4^{-2}$, H^+ for the balanced reaction respectively are

(1) 2, 5, 16 (2) 16, 3, 12

(3) 15, 16, 12 (4) 2, 16, 5

75. Which of the following bonding is not possible :

(1) $p_x + p_x - \sigma$ bond along x-axis

(2) $p_z + d_{xz} - \pi$ bond along x-axis

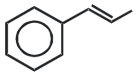
(3) $p_y + p_y - \pi$ bond along x-axis

(4) $d_{yz} + d_{yz} - \pi$ bond along x-axis

76. Which of following does not show zero overlapping (z-axis is inter nuclear axis) :

(1) $s + p_x$ (2) $p_x + p_y$

(3) $p_y + s$ (4) $p_y + p_y$

77. How many bonds are there in  :-

(1) 13 (2) 23

(3) 20 (4) 26

78. The hybridisation of S in SF_4 is:

(1) dsp^2 (2) sp^3

(3) sp^3d (4) sp^3d^2

79. Which of the following consist of equal no. of $p\pi-p\pi$ & $p\pi-d\pi$ bonds :-

(1) SO_3 (2) SO_2

(3) CO_2 (4) SO_4^{-2}

80. Incorrect statement is :-

(1) Bond form by s−s overlapping is always weakest

(2) SiO_2 is 3-D network solid

(3) Coaxial overlapping always produce σ bond

(4) Hybrid orbitals produce σ bond

81. Existence of which of the following species is not possible ?

(1) SiF_6^{-2} (2) $SiCl_6^{-2}$

(3) $GeCl_6^{-2}$ (4) SCl_2

82. Which of the following molecule is linear?

(1) PCl_5 (2) SF_4

(3) ClF_3 (4) XeF_2

83. sp^3d^2 Hybridization is not shown by :-

(1) BrF_5 (2) SF_6

(3) XeF_4 (4) PF_5

84. Incorrect statement for protonation of NH_3 is :-

(1) Co-ordinate bond is formed

(2) Bond angle increases

(3) NH_3 act as a lewis base

(4) %s-character in hybrid orbital increases

85. Correct order of size is :-

(1) F > C > Br > Ge

(2) Ge > C > F > Br

(3) Ge > Br > C > F

(4) Ge > C > Br > F

86. Which two are closest to one another in size ?

(1) Li^+ and Na^+ (2) Be^{+2} and Mg^{+2}

(3) Be^{+2} and Li^+ (4) Li^+ and Mg^{+2}

87. If Z_{eff} of sodium is 'x' then Z_{eff} of calcium is :-

(1) x + 0.35 (2) x + 0.65

(3) x – 0.35 (4) x – 0.65

88. Suppose an orbital may accomodate 3 electrons then the number of elements in IV period :-

(1) 18 (2) 27

(3) 12 (4) 45

89. Which of the following statement is not correct?

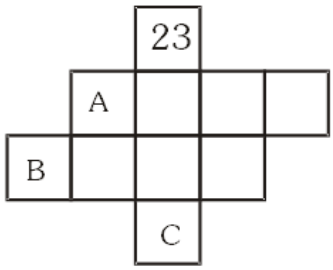
(1) Tc and Pm are not found in nature

(2) Element having ns^2 np^5 configuration are placed in the column before, the extreme right in the periodic table.

(3) Most of the man made element occurs in the actinoid series

(4) The 3d - transition series contains elements having atomic number from 21 to 29

90. Which of the following statement is not correct?



(1) Atomic no. of A, B, C are 40, 57, 105 respectively.

(2) Group no of A, B, C are IV B, III B and V B respectively

(3) Period no of A, B, C are, 4th, 5th and 6th resp.

(4) C is a radioactive element

SUBJECT : BIOLOGY

Topic : CELL BIOLOGY COMPLETE, LIVING WORLD, BIOLOGICAL CLASSIFICATION, PLANT KINGDOM, ANIMAL TISSUE, COCKROACH, FROG, BIOMOLECULES, BEATHING EXCHANGE OF GASES .

91. Which of the following statements is **wrong** ?

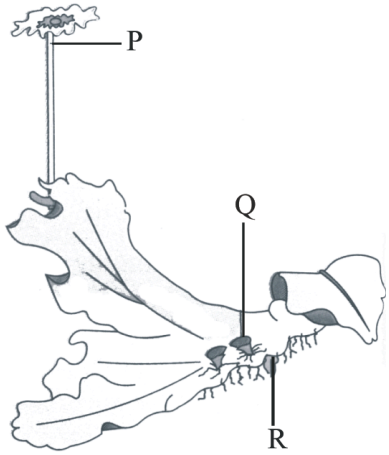
(1) Capsule is the part of sporophyte of Bryophyta

(2) Basidiocarp is the fruiting body of fungus.

(3) Endosperm are produced by pteridophyta.

(4) Naked seeds are seen in Gymnosperms.

92. In given figures, identify P, Q and R :-



	P	Q	R
(1)	Antheridiophore	Gemma cup	Rhizoids
(2)	Antheridal branch	Capsule	Archegoniophore
(3)	Archegoniophore	Capsule	Rhizoids
(4)	Antheridiophore	Gemma cup	Roots

93. (a)- Stem of *Pinus* and *Cedrus* are branched
(b)- *Selaginella* is heterosporous pteridophyte
(c)- Mustard is a flowering plant
(d)- *Marchantia* is dioecious bryophyta
(e)- Protonema is the character of Moss.
From given correct statement is/are .....

(1) Only a, c, d

(2) Only a, b, c

(3) Only b, c, d

(4) All are correct

94. How many members belong to phycomycetes list given below ?
Albugo, Penecillium, Neurospora*, Volvox, Mucor, Fusarium, Ustilago, Rhyzopus, .*

(1) Seven (2) Five (3) Four (4) Three

95. Match the column I with column II and choose correct option :-

	Column I		Column II
(i)	Red algae	(a)	Pyrenoids
(ii)	Brown algae	(b)	Agar - Agar
(iii)	Blue green algae	(c)	Alginate
(iv)	Green algae	(d)	Water blooms

(1) (i) – (d), (ii) – (c), (iii) – (b), (iv) – (a)

(2) (i) – (b), (ii) – (c), (iii) – (d), (iv) – (a)

(3) (i) – (a), (ii) – (b), (iii) – (d), (iv) – (c)

(4) (i) – (b), (ii) – (a), (iii) – (d), (iv) – (c)

96. Which of the following is a correct statement ?

(1) Male and female gametophyte is independant in angiosperm and gymnosperms

(2) Two unequal flagella are present in dinoflagellata and euglenoids

(3) Diatoms show flagellary movement

(4) Virus, viroid and prions cause disease in plants and animals both

97. Which of the followings is/are symbiotic associations between autotrophic and heterotrophic organisms only
(A) Mycorrhiza between fungi and roots of higher plants
(B) Lichen between phycobiont and mycobiont.
(C) Coralloid roots of Cycas having cyanobacteria
(D) Azolla and Anabena

(1) A & B (2) B & C

(3) A, B & C (4) A, B & D

98. Which one is a prokaryote ?

(1) Oscillatoria (2) Chlamydomonas

(3) Volvox (4) Chlorella

99. Which one of the following show isogamy with flagellated gametes ?

(1) *Volvox* (2) *Chara*

(3) *Ulothrix* (4) *Polysiphonia*

100. According to five kingdom classification system, below organisms of groups belong to how many kingdoms?
Chlorella, Chlamydomonas, Chara, Euglena, *Cycas, Pinus,* Mango, *Amoeba, Nostoc, Plasmodium,* Mycoplasma, Slime moulds

(1) 4 (2) 3 (3) 2 (4) 1

101. Chlorophyll common between Phaeophyceae and Diatoms but absent in Rhodophyceae is

(1) Chlorophyll a

(2) Chlorophyll c

(3) Chlorophyll b

(4) Chlorophyll e

102. If a plant group is classified on the basis of chromosome number, chromosome structure etc. then the classification is based on?

(1) Chemotaxonomy

(2) NOR of chromosomes

(3) Cytotaxonomy

(4) All

103. Two kingdom system of classification is mainly based on:-

(1) Cell wall

(2) Cell membrane

(3) Mode of nutrition

(4) Phylogeny

104. In slime moulds :-
(a) The main body is wall less
(b) True wall is found around spores
(c) Spores are dispersed by water
(d) Saprophytes are not common

(1) a, b are correct

(2) b, d are correct

(3) a, c are correct

(4) a, d are correct

105. How many chromosomes shall be present in a diploid cell at mitotic anaphase, if ovum of same species has 40 chromosomes.

(1) 20 (2) 40

(3) 160 (4) 120

106. If the n = 32 in plant cell, then what is possible in metaphase-I of meiosis

(1) 32 bivalents, 64 centromere, 128 chromatids

(2) 16 tetravalents, 16 centromere, 128 chromatids

(3) 16 bivalents, 32 centromere, 128 chromatids

(4) 34 bivalents, 32 centromere, 128 chromatids

107. Red tide is caused by :-

(1) Rapid multiplication of red algae

(2) Rapid multiplization of euglenoids

(3) Rapid multiplication of slime molds

(4) Rapid multiplication of dinoflagellates

108. Choose the correct sequence.

(1) Meiosis $\longrightarrow$ Plasmogamy $\longrightarrow$ Karyogamy $\longrightarrow$ dikaryon

(2) Plasmogamy $\longrightarrow$ dikaryon $\longrightarrow$ karyogamy $\longrightarrow$ meiosis

(3) Plasmogamy $\longrightarrow$ karyogamy $\longrightarrow$ meiosis $\longrightarrow$ dikaryon

(4) Plasmogamy $\longrightarrow$ meiosis $\longrightarrow$ karyogamy $\longrightarrow$ dikaryon

109. Match the column -

	Column-I		Column-II
(i)	Virus	(a)	Composite organism
(ii)	Viroids	(b)	Protein enclosing nucleic acid
(iii)	Prions	(c)	Abnormally folded protein
(iv)	Lichen	(d)	Low molecular RNA.

(1) i-b, ii-c, iii-d, iv-a (2) i-b, ii-d, iii-a, iv-c

(3) i-b, ii-d, iii-c, iv-a (4) i-c, ii-b, iii-d, iv-a

110. How many of the following are sexual spores in fungi ?

Conidia, sporangiospore, zoospore, ascospore, basidiospore, oospores

(1) 1 (2) 2 (3) 3 (4) 4

111. How many of the following are heterosporous in nature-

Sphagnum, Cedrus, Selaginella, Salvinia, Sargassum, Azolla, Dryopteris, Equisetum, Cycas

(1) 3 (2) 4 (3) 5 (4) 7

112. **Assertion :-** *Neurospora* is used extensively in biochemical and genetic work.

Reason :- *Neurospora* growfast, easily cultivate and short life span.

(1) Both Assertion and Reason are true and Reason is correct explanation of the Assertion.

(2) Both Assertion and Reason are true but R is not correct explanation of the Assertion.

(3) Assertion is true and Reason is false.

(4) Assertion and Reason both are false

113. Cell wall is absent in :-

(1) *Nostoc* (2) *Anabaena*

(3) Mycoplasma (4) *E. Coli*

114. Select the total number of families from the following-

Felis, Felidae, Solanaceae, Liliaceae, Canidae, Hominidae Poaceae, Muscidae, Insecta Convolulaceae

(1) 5 (2) 6 (3) 7 (4) 8

115. Arrange the events of interphase in the **correct** sequences

(i) RNA synthesis, DNA polymerase synthesis

(ii) Replication of DNA histone protein synthesis

(iii) Tubulin protein synthesis, ATP synthesis

(iv) Karyokinesis, cytokinesis

(1) (i) ⟶ (ii) ⟶ (iii) ⟶ (iv)

(2) (ii) ⟶ (i) ⟶ (iii) ⟶ (iv)

(3) (iv) ⟶ (iii) ⟶ (ii) ⟶ (i)

(4) (iii) ⟶ (iv) ⟶ (ii) ⟶ (i)

116. Arrange the following events of meiosis in correct sequence:

A. Crossing over

B. Synapsis

C. Terminalisation of chiasmata

D. Disappearance of nucleolus

(1) A, B, C, D (2) B, C, D, A

(3) C, A, B, D (4) B, A, C, D

117. X shaped structure on chromosomes is the characteristics of.

(1) Mitosis (2) Meiosis-I

(3) Meiosis-II (4) Both (2) and (3)

118. Which is correct about quiescent stage of cell.

(1) G_0–phase, cell metabolically passive

(2) G_1–phase, cell metabolically active

(3) G_0–phase, cell metabolically active

(4) G_1–phase, cell metabolically passive

119. Transport of male gametes in bryophytes and pteridophytes occurs through-

(1) Wind (2) Water

(3) Birds (4) Insects

120. In a diploid cell after S-phase quantity of DNA is 20 pico gram then after anaphase-II situation will be in each daughter nucleus :-

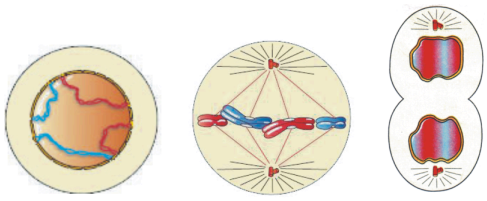
(1) Diploid, 10 pico gram DNA

(2) Haploid, 10 pico gram DNA

(3) Haploid, 5 pico gram DNA

(4) Diploid, 5 pico gram DNA

121. The figures below show three stages (A, B, C) of mitotic division. Select the option giving correct identification :-



(A) (B) (C)

(1) B-Individual chromosomes can no longer be seen.

(2) A-Initiation of condensation of chromosomal material.

(3) C-Initiation of assembly of mitotic spindle.

(4) B-Chromosomes consist of one chromatid.

122. Which one of the following event is not related with meiosis-I :-

(1) Segregation of homologous chromosomes

(2) Crossing over between non sister chromatids

(3) Sepration of sister chromatids

(4) Condensation of chromatin

123. Crossing over occurs in.

(1) Pachytene (2) Zygotene

(3) Leptotene (4) Diplotene

124. Which of the following cytoskeleton protein help in cytokinesis in animal cells

(1) Microtubule

(2) Microfilament

(3) Intermediate filament

(4) Tubulin protein

125. Which of the following statement is incorrect about the plasma membrane of RBC

(1) 52% protein and 40% lipids are in the membrane of RBC

(2) Head of lipid is hydrophilic

(3) Integral proteins are found on the external surface of plasma membrane

(4) Tail of lipid in hydrophobic

126. Which one is the continuous with outer membrane of Nucleus :-

(1) Golgibody

(2) Mitochondria

(3) RER

(4) Lysosome

127. Match the following column correctly:-

	Column-I		Column-II
(a)	Centriole	(i)	Infolding of inner mitochondrial membrane
(b)	Chlorophyll	(ii)	Thylakoid
(c)	Cristae	(iii)	E.R.
(d)	Cisternae	(iv)	Basal body of cilia & flagella

(1) a-iv, b-ii, c-i, d-iii (2) a-iv, b-ii, c-iii, d-i

(3) a-iv, b-iii, c-ii, d-i (4) a-iv, b-i, c-ii, d-iii

128. **Assertion (A):** Centrosome does not show any kind of compartmentalisation of cell cytoplasm.
Reason (R): Centrosome is devoid of membranous coverings.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

129. If nuclear pores are blocked then movement of which of the following will be affected :-

(1) DNA and RNA

(2) DNA and proteins

(3) Only proteins

(4) RNA and proteins

130. At anaphase sub-metacentric chromosomes look like :-

(1) V - shaped (2) L - shaped

(3) rod shaped (4) i - shaped

131. Which of following is not a defining feature of living organism?

(1) Reproduction

(2) Metabolism

(3) Consciousness

(4) Cellular organisation

132. Cell division in animals occur ___A___ and in plants occur ___B___. What is A and B ?

(1) A : Throughout life, B : Upto a certain age

(2) A : Throughout life, B : Throughout life

(3) A : Upto a certain age, B : Thruoughout life

(4) A : Upto a certain age, B : Upto a certain age

133. **Assertion :** In fungi, vegetative reproduction occurs by fragmentation and budding.
Reason : Asexual reproduction in fungi, occurs through formation of asexual spores.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

134. The protonema is a stage in the life cycle of

(1) Riccia

(2) Funaria

(3) all bryophytes

(4) Pinus

135. If a root tip cell contain 40 pg of DNA and 20 chromosome then after S-phase the cell will have :

(1) 40 pg of DNA and 20 chromosome

(2) 80 pg of DNA and 10 chromosome

(3) 40 pg of DNA and 10 chromosome

(4) 80 pg of DNA and 20 chromosome

136. In the given table choose the correct value of partial pressure :-

Respiratory gas	Alveoli	Deoxygenated blood	Oxygenated blood	Tissue
O_2	104	b	95	d
CO_2	a	45	c	45

	a	b	c	d
(1)	40	40	40	40
(2)	45	45	95	45
(3)	45	40	45	95
(4)	40	95	40	45

137. Read the following statements.

(a) High $H^{\oplus}$ concentration and higher temperature are favourable for dissociation of oxygen from oxyhaemoglobin.

(b) Partial pressure of oxygen in alveoli is higher than in Tissue.

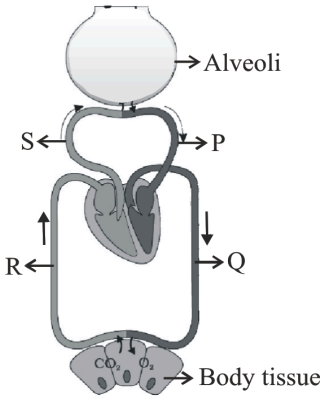
(c) The maximum volume of air a person can breathe in after a forced expiration is vital capacity

(d) In alveoli low PCO_2 and high PO_2 are responsible for dissociation of CO_2 from carbamino-haemoglobin.

How many statements are correct ?

(1) One (2) Two (3) Three (4) Four

138. Given below the diagram represents of exchange of gases at alveolar level and tissue level in which some parts are labelled by P, Q, R & S. Find out the correct lebelling with regards to partial pressure of gases in blood ?



(1) Q = Systemic artery in which P_{O_2} is lesser as compare to 'S'.

(2) S = Pulmonary vein in which P_{O_2} & P_{CO_2} are 95 & 45 mmHg respectively.

(3) R = carries deoxygenated blood in which P_{CO_2} is more while P_{O_2} is less as compare to 'P'.

(4) P = carries oxygenated blood in which P_{O_2} and P_{CO_2} are similar to blood of 'R'.

139. Which of the following volume or capacity of lungs can't be measured directly by the spirometer ?

(1) Residual volume

(2) Total lung capacity

(3) Functional residual capacity

(4) All of the above

140. A child was born alive but was killed by a man. He told the mother later it was a still born. The mother doubted his statement and case was reported to police. Post mortem report confirmed that it was a murder. The test performed by the medical expert was taking a piece of lung and putting in water where it :-

(1) Kept floating

(2) Settled at the bottom

(3) Showed blood clots

(4) None of these

141. Dissociation curve shift to left when there is:-

(1) Decrease in O_2 concentration

(2) Decrease in CO_2 concentration

(3) Increase in CO_2 concentration

(4) Increase in 2-3-DPG/BPG concentration

142. What will be the P_{O_2} and P_{CO_2} in the atmospheric air compared to those in the alveolar air ?

(1) P_{O_2} lesser, P_{CO_2} higher

(2) P_{O_2} higher,P_{CO_2} lesser

(3) P_{O_2} higher,P_{CO_2} higher

(4) P_{O_2} lesser, P_{CO_2} lesser

143. The conditions which are favourable for the formation of oxyhaemoglobin are :-

(1) $\downarrow PO_2$, $PCO_2\uparrow$, H^+ concentration $\downarrow$, Temperature$\downarrow$

(2) $PO_2\downarrow$, $PCO_2\downarrow$, H^+ concentration $\uparrow$, Temperature$\uparrow$

(3) $\uparrow PO_2$, $PCO_2\downarrow$, H^+ concentration $\downarrow$,Temperature$\downarrow$

(4) $\downarrow PO_2$, $PCO_2\uparrow$,H^+ concentration$\downarrow$, Temperature$\uparrow$

144. How many cellular layers are present in the diffusion membrane in lungs ?

(1) One (2) Two (3) Three (4) Four

145. Correct sequence of layers of diffusion membrane from blood capillary to alveolar wall is :-

(1) Thin squamous epithelium of alveoli, endothelium of alveolar capillaries and basement substance respectively

(2) Endothelium of alveolar capillaries, basement substance and thin squamous epithelium of alveoli respectively.

(3) Endothelium of alveolar capillaries, thin squamous epithelium of alveoli and basement substance respectively.

(4) Thin squamous epithelium of alveoli basement substance and endothelium of alveolar capillaries respectively.

146. **Statement-I :** A healthy man can inspire or expire approximately 6000 to 8000 ml of air per minute.
Statement-II : Additional volume of air, a person can inspire by a forcible inspiration is called inspiratory reserve volume.

(1) Both Statement are correct

(2) Both Statement are incorrect

(3) Statement-I is correct while Statement-II is incorrect

(4) Statement-I is incorrect while Statement-II is correct

147. **Assertion :** Thickness of Respiratory membrane is less than 1 mm.
Reason : Thinner membrane increases diffusion rate.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

148. **Assertion :** Pneumotaxic centre increases respiration rate.
Reason : Neural signals from pneumotaxic centre reduces the duration of inspiration.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

149. **Assertion (A) :-** Contraction of diaphragm causes increase in thoracic volume anterio-posteriorly.
Reason (R) :- Due to contraction diaphragm move down and become flatten.

(1) A and R both are correct and R is correct explanation of A

(2) A and R both are correct but R is not correct explanation of A

(3) A correct but R incorrect

(4) A incorrect but R is correct

150. Select the incorrect statement regarding polysaccharide?

(1) Glucose is a monomer of cellulose.

(2) Both starch and glycogen are present as a store house of energy in plant tissue.

(3) Inulin is a polymer of fructose.

(4) Monosaccharide unit of chitin is N–acetyl glucosamine.

151. Which of the following secondary metabolites used in drugs?

(1) Vinblastine and curcumin

(2) Anthocyanin

(3) Gums and cellulose

(4) Abrin and ricin

152. How many carbohydrates in the list given below are polysaccharide in nature :-
Cellulose, Starch, Glycogen, Chitin, inulin, Glucose, Maltose, Sucrose

(1) 5 (2) 4 (3) 6 (4) 2

153. Match the column I with column II

	Column I		Column II
A	Phospholipid	i	β - D Glucose polymer
B	Chitin	ii	α - D Glucose polymer
C	Glycogen	iii	N - acetyl glucosamine
D	Cellulose	iv	Lecithin

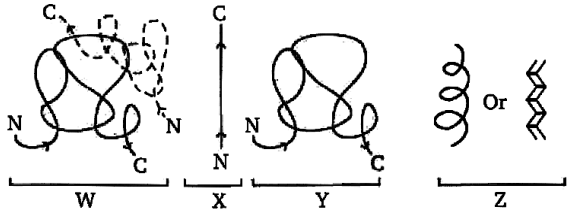
(1) A - iv, B - iii, C - ii, D - i

(2) A - i, B - ii, C - iii, D - iv

(3) A - ii, B - iii, C - i, D - iv

(4) A - iv, B - i, C - ii, D - iii

154. Find out the primary, secondary, tertiary and quaternary protein structures in the diagram :



	W	X	Y	Z
(1)	Quaternary	Secondary	Primary	Tertiary
(2)	Tertiary	Primary	Quaternary	Secondary
(3)	Quaternary	Primary	Tertiary	Secondary
(4)	Secondary	Primary	Quaternary	Tertiary

155. Consider the following four statements (a-d) and select the option which includes all the correct ones only :-

(a) Lipids are generally water insoluble.

(b) The R group could be a methyl ($-CH_3$) or ethyl ($-C_2H_5$) or higher no. of $-CH_2$ group (1 carbon to 19 carbon) in fatty acids

(c) Palmitic acid has 16 carbons including the carboxyl carbon.

(d) Arachidonic acid has 21 carbon atom including the carboxyl carbon.

Option

(1) Statement (a), (b) and (c)

(2) Statement (a), (b) only

(3) Statement (c), (d) only

(4) Statement (a), (c) and (d)

156. A typical fat molecule is made up of :-

(1) Three glycerol molecules and one fatty acid molecule

(2) One glycerol and three fatty acid molecules

(3) One glycerol and one fatty acid molecule

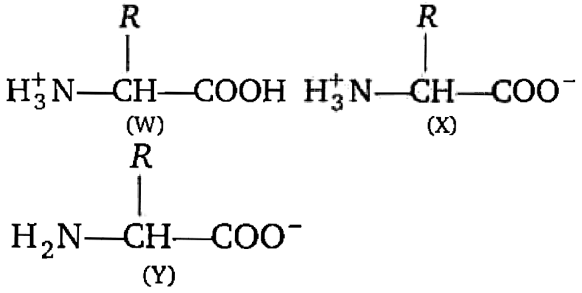
(4) Three glycerol and three fatty acid molecules

157. Which of the following disaccharide has minimum one glucose unit ?

(1) Maltose (2) Lactose

(3) Sucrose (4) All of the above

158. Which of the following is a zwitter ion ?



(1) W (2) W and X

(3) W and Y (4) X

159. A Tetrasaccharide made up of four monosaccharide units; A, B, C & D. If their molecular weight is respectively 120, 180, 210, 230, then what will be the M_w of that tetrasaccharide ?

(1) 740 (2) 704

(3) 686 (4) 722

160. Read the following (A-D) statements :-

(A) Connective tissue are most abundant and widely distributed in the body of complex organism.

(B) Connective tissue helps in linking and supporting other tissue or organs in the body.

(C) Adipose tissue is a type of dense connective tisue.

(D) Connective tissue does not secrete its matrix.

Choose the incorrect statements :

(1) A, C & D (2) B and D

(3) C and D (4) Only D

161. Male cockroach and female cockroach can be differentiated by :

(1) Ocelli (2) Anal cerci

(3) Anal styles (4) Head

162. The excess of nutrients which are not used immediately are converted into fats are stored in :

(1) Areolar tissue

(2) Adipose tissue

(3) Dense regular connective tissue

(4) Dense irregular connective tissue

163. Connective tissue cell also secrete modified __(A)__, which accumulate between cells and fibres and act as __(B)__ ?

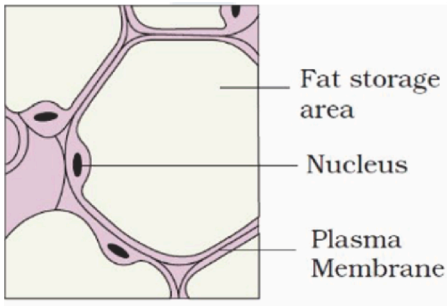
(1) (A) Matrix, (B) Polysaccharides

(2) (A) Matrix, (B) Oligosaccharides

(3) (A) Polysaccharides, (B) Matrix

(4) (A) Oligosaccharides, (B) Matrix

164. Identify the given below diagram and select the correct option ?



(1) Areolar tissue : It serves as a supportive frame form for epithelium

(2) Adipose tissue : It is another type of dense connective tissue located mainly beneath the skin.

(3) Areolar tissue : The cells of this tissue are specialised to store fats.

(4) Adipose tissue : it is another type of loose connective tissue located mainly beneath the skin.

165. Match the column and find out the correct answer.

	Column-I		Column-II
A	Most abundant formed elements of blood	(i)	Plateles
B	Collagen of areolar tissue is produced by	(ii)	Mast cells
C	Tendons are composed of this tissue	(iii)	Dense regular
D	Provide shape to external ear	(iv)	Elastic cartilage
		(v)	Hyaline cartilage
		(vi)	Erythrocytes
		(vii)	Fibroblast
		(viii)	Areolar tissu

(1) A-i B-viii C-iii D-v

(2) A-vi B-ii C-iii D-viii

(3) A-i B-vii C-viii D-iv

(4) A-vi B-vii C-iii D-iv

166. Match the column-I and Column-II :-

Column-I		Column-II	
(I)	Mast cell	(A)	Histamine
(II)	Fibroblast	(B)	Antibody
(III)	Lymphocyte	(C)	Phagocytosis
(IV)	Macrophage	(D)	Fibres

(1) I - A, II - B, III - D, IV - C

(2) I - A, II - D, III - B, IV - C

(3) I - A, II - D, III - C, IV - B

(4) I - A, II - B, III - C, IV - D

167. The forebrain of frog includes _____ complete the above statements -

(1) Olfactory lobes, cerebral hemisphere, and unpaired diencephalon

(2) Optic lobes and diencephalon

(3) Cerebellum and Medulla oblongata

(4) Cerebral hemisphere and cerebellum

168. Metamorphosis in frog can be promoted by-

(1) Iodine (2) Calcium

(3) Chlorine (4) Phosphorus

169. Consider the following four statements (A-D) related to the common frog *Rana trigrina*, and select the correct option stating which ones are true (T) and which ones are false (F).

(A) On dry land, it would die due to lack of O_2 if its mouth is forcibly kept closed for a few days.

(B) It has a four-chambered heart.

(C) On dry land, it turns uricotelic from ureotelic.

(D) Eyes are covered with nictitating membrane.

(1) (A)–F, (B)–T, (C)–T, (D)–F

(2) (A)–T, (B)–F, (C)–F, (D)–T

(3) (A)–T, (B)–T, (C)–F, (D)–F

(4) (A)–F, (B)–F, (C)–T, (D)–T

170. In frog a triangular structure called ___A__ joins the __B__.

(1) A - conus arteriosus, B - Left atrium

(2) A - sinus venosus, B - ventricle

(3) A - sinus venosus, B - right atrium

(4) A - conus arteriosus, B - right atrium

171. RBC's of Frog contain–

(1) Nucleus but not haemoglobin

(2) Haemoglobin but not nucleus

(3) Both nucleus and haemoglobin

(4) Plasma proteins

172. Select the incorrect statement w.r.t. Frog–

(1) Protective colouration in frog is called mimicry

(2) Ventral side of skin is generally olive green

(3) Final digestion takes place in the intestine

(4) Only male possess sound producing prominent vocal sacs

173. At a time mature femal frog can lay...ova at a time:

(1) 2000-2500 (2) 2500-3000

(3) 1000-2000 (4) 1500-2000

174. For female cockroaches, how many statements are **correct**?

(a) The 7th sternum is boat shaped

(b) The 7th sternum together with the 8th and 9th sterna forms a brood or genital pouch

(c) The anterior part of genital pouch contains female genopore, spermathecal pores and collateral glands

(d) Only right collateral gland form ootheca

(1) a and b (2) b and d

(3) a, b, c (4) a and d

175. Given below are two statements :

Statement-I :- Malpighian tubules are thin yellow coloured filamentous structures at the junction of foregut & midgut.

Statement-II :- Malpighian tubules help in osmoregulation and excretion.

Choose the most appropriate answer from the option given below

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

176. Which of the following statement is correct about cockroach?

(1) Blood vascular system of cockroach is an open type

(2) Excretion is performed by tracheal tube

(3) Respiration is performed by malpighian tubules

(4) Excretory matter is NH_3

177. What is the correct sequence of organs in the digestive system of cockroach?

(a) Oesophagus (b) Ileum
(c) Gizzard (d) Colon
(e) Rectum (f) Crop

Options:

(1) a, f, c, d, e, b

(2) a, f, c, e, d, b

(3) a, f, c, b, d, e

(4) a, c, f, d, e, b

178. On average how many oothecae are produced by female cockroach :

(1) 7–8

(2) 9–10

(3) 12–13

(4) 19

179. In cockroach, mouth parts are of -

(1) Chewing and lapping

(2) Siphoning type

(3) Piercing and sucking type

(4) Biting and chewing type

180. Choose the **correct** statements :

(a) Bones support and protect softer tissues and organs.

(b) Weight bearing function is served by limb bones.

(c) Ligament is the site of production of blood cells.

(d) Adipose tissue is specialised to store fats.

(e) Tendons attach one bone to another.

Choose **the most appropriate answer** from the options given below :

(1) (a), (b) and (d) only

(2) (b), (c) and (e) only

(3) (a), (c) and (d) only

(4) (a), (b) and (e) only

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